

Ahmad Amine

Philadelphia, PA

ahmadamine998.github.io | github.com/AhmadAmine998

EDUCATION

University of Pennsylvania

Ph.D., Electrical and Systems Engineering

Advisor: Rahul Mangharam

M.S.E., Robotics; GPA: 3.91/4.0

Philadelphia, PA

Aug 2023–Present

Expected graduation: May 2028

May 2023

American University of Beirut

B.E., Electrical and Computer Engineering; Minor in Mathematics

Beirut, Lebanon

May 2020

RESEARCH INTERESTS

Safe learning-based control; model predictive control; learning from failures and successes; online system identification; autonomous vehicles.

SELECTED PUBLICATIONS

* Equal contribution.

Journal and Conference Papers

A. Amine, N.-M. T. Kokolakis, U. Rosolia, T. X. Nghiem, R. Mangharam. Failure-Aware Iterative Learning of State-Control Invariant Sets. *IEEE Conference on Decision and Control (CDC)*, 2026 (accepted).

N. T. Nguyen*, B. Nguyen*, **A. Amine**, T. Vo-Duy, R. Mangharam, T. X. Nghiem. AD-MPCC: Adaptive Differentiable Model Predictive Contouring Control for Autonomous Racing. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026 (accepted).

H. Zheng, Z. Zang, **A. Amine**, C. I. Vasile, R. Mangharam. STL-SVPIO: Signal Temporal Logic Guided Stein Variational Path Integral Optimization. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026 (accepted).

A. Amine, K. Puri, V.-A. Le, R. Mangharam. Nonplanar Model Predictive Control for Autonomous Vehicles with Recursive Sparse Gaussian Process Dynamics. *IEEE Intelligent Vehicles Symposium (IV)*, pp. 2088–2093, 2026.

Z. Zang*, **A. Amine***, N.-M. T. Kokolakis, T. X. Nghiem, U. Rosolia, R. Mangharam. SIT-LMPC: Safe Information-Theoretic Learning Model Predictive Control for Iterative Tasks. *IEEE Robotics and Automation Letters*, 11(1):986–993, 2026.

T. Nagy, **A. Amine**, T. X. Nghiem, U. Rosolia, Z. Zang, R. Mangharam. Ensemble Gaussian Processes for Adaptive Autonomous Driving on Multi-friction Surfaces. *IFAC-PapersOnLine*, 56(2):494–500, 2023.

Workshop Papers

R. Tumu*, **A. Amine***, L. Milburn*, R. Gupta, U. Kono, R. Mangharam. Zero-Shot Context Identification through Clustering and Foundation Modeling for Friction Estimation. *ICRA Workshop on Foundation Models and Neuro-Symbolic AI for Robotics*, 2025.

Z. Zang, **A. Amine**, R. Mangharam. PoseINN: Realtime Visual-based Pose Regression and Localization with Invertible Neural Networks. *RoboNerF: 1st Workshop on Neural Fields in Robotics, ICRA*, 2024.

RESEARCH EXPERIENCE

Research Assistant | mLab/xLab, University of Pennsylvania

Jun 2022–Present

Philadelphia, PA

- Developed FAIL to learn maximal state-control invariant sets from failing trajectories, with monotonic convergence guarantees for deterministic linear time-invariant systems under polytopic constraints.
- Developed nonplanar model predictive control with recursive sparse Gaussian-process dynamics for autonomous vehicles operating on uneven terrain.
- Reduced model-update time for a 40-step MPC horizon from 55 ms to 55 μ s (1,000 $\times$) using SciPy sparse matrices.
- Reduced CVXPYgen controller solve time from 20 ms to 7 ms (approximately 3 $\times$).
- Implemented ensemble Gaussian-process MPC with learned model weights for real-time adaptation to changing driving conditions, achieving a 10 $\times$ speedup over the lab baseline.
- Proposed and implemented an adaptive penalty method for safe model predictive path integral control (MPPI), with real-time deployment on NVIDIA Jetson AGX hardware.
- Co-developed Roboracer R5, an open-source 1/5-scale off-road robot, integrating its power system and Fixposition Vision-RTK.
- Deployed SIT-LMPC and friction-estimation methods on Roboracer R5.
- Trained an MPPI auto-tuner with PPO to improve tracking under time-varying dynamics.
- Developed an Isaac Sim vehicle simulator for off-road control experiments.

Research Assistant | Vision and Robotics Lab, American University of Beirut

Jun 2020–Aug 2021

Beirut, Lebanon

- Developed a bidirectional cloud communication architecture using AWS to connect robots with virtual assistants.
- Built a ROS/Gazebo simulation environment with custom robot models created using Blender and URDF.

INDUSTRY EXPERIENCE

Controls Intern | Tier IV

Jul–Aug 2023

Tokyo, Japan

- Developed a vehicle steering system-identification pipeline with ROS bag preprocessing and ARX–Gaussian process models, achieving 97.1% NRMSE-based validation fit in MATLAB.
- Designed identification experiments with clothoid trajectories and constrained nonlinear MPC in CasADi.
- Deployed learned models as a simulation-based digital twin to support control-stack integration.
- Reformulated MPC in Autoware’s obstacle-avoidance planner to exploit sparsity in OSQP, reducing computation time.

Mechatronics Engineer | Band Industries

Jun 2020–Aug 2021

Matn, Lebanon

- Developed RTOS firmware for ARM and Xtensa LX6 devices, including drivers and system integration for audio and control products.
- Rewrote the Bluetooth Low Energy protocol in C/C++ and introduced regression testing with GoogleTest and test-driven development.
- Eliminated motor encoders through open-loop position control, meeting performance specifications and reducing hardware costs by 5%.
- Implemented telemetry, eMMC logging, HTTPS over-the-air updates and the Roadie Coach user interface in C with LVGL.

TEACHING EXPERIENCE

University of Pennsylvania

Lead Teaching Assistant, ESE 6150: F1/10 Autonomous Racing Cars Spring 2024–2026

- Delivered lectures on optimal control, model predictive control, stochastic optimal control and reinforcement learning.

Teaching Assistant, MEAM 5200: Introduction to Robotics Spring 2022–2023; Fall 2023

- Developed an AprilTag vision system for Franka Panda arms, including emulated cameras in Gazebo.

Teaching Assistant, ESE 5000: Linear Systems Fall 2024

Teaching Assistant | American University of Beirut

EECE 311: Electronic Circuits Spring 2019

EECE 310: Electronics Fall 2018

EECE 230: Introduction to Computation and Programming Spring 2018

OPEN-SOURCE CONTRIBUTIONS AND LEADERSHIP

Developer and Maintainer | F1TENTH Ecosystem Jan 2024–Present

- Maintain simulation and control repositories, including issue triage, pull-request review, integration and releases.

f1tenth_system | f1tenth_planning | f1tenth_gym

Organizer | Roboracer (F1TENTH) Competitions Jan 2024–Present

- Organized ICRA 2026 (200 participants, 30 teams from more than 12 countries) and IV 2026 (14 teams).
- Previously organized ICRA 2025 (26 teams), CDC 2024 (12 teams) and ICRA 2024 (18 teams).
- Organizing the upcoming IROS 2026 competition.
- Coordinate race setup, competition rules, website, event schedule, live operations and Twitch streaming.

HONORS AND AWARDS

Joseph A. Ganster, Jr. and Julia A. Ganster Engineering Fellow Apr 2023

ESE Endowed Fellowship, University of Pennsylvania

Dean's Fellowship, University of Pennsylvania Apr 2023

LIFE Scholarship, LIFE Lebanon Aug 2021

PROFESSIONAL SERVICE

Peer Reviewer Aug 2023–Present

Journals: IEEE Robotics and Automation Letters; Automatica; Optimization and Engineering.

Conferences and symposia: ICRA (2025–2026), IROS (2023–2026), RO-MAN (2023–2026), IFAC SYSID (2024), COINS (2025), IV (2026).

TECHNICAL SKILLS AND LANGUAGES

Programming: Python, C/C++, MATLAB, Simulink

Control: MPC, NMPC, MPPI, LMPC

Optimization: CasADi, CVX, CVXPY, CVXPYgen, OSQP, mpls

Machine learning: JAX, Flax, PyTorch, GPyTorch, GPJax, BoTorch

Robotics and development: ROS 1/ROS 2, Gazebo, Isaac Sim, Linux, Docker, Git

Languages: Arabic (native), English (fluent), Japanese (basic)